Unlocking the Secrets of Bitcoin's Inner Workings: The Math,

In ***Bitcoin***, **functions** are utilized to **execute transactions**. We already know that transactions in ***Bitcoin*** are essentially transfers of value from one address to another. To perform a transaction, a number of cryptographic functions are used to validate the transaction and update the state of the ***Bitcoin*** blockchain, which is a decentralized ledger that keeps track of all transactions.

The functions used in a ***Bitcoin*** transaction include verifying the authenticity of the transaction inputs, checking that the sender has sufficient funds, and updating the balances of the relevant addresses. Once a transaction is verified and added to the blockchain, it becomes part of the permanent record of all transactions on the network.

- A **one-way function** uses a set of instructions to process the information and turns it into something **new**, like a smoothie recipe turns ingredients into a new drink. But, just as **you can't un-blend a smoothie** to get the original ingredients back, **you can't reverse the one-way function to get the original information back.**

Public-key cryptography, of which **public key** is a part of, relies on the use of **one-way functions**, which make it difficult to determine the **private key** from the **public key**. It is not exactly "impossible" to find the private key from the **public key**, but it is extremely difficult to do so, and it would take an inordinate amount of time and computational power to accomplish this task.

- Finding a **private key** from a **public key** in ***Bitcoin*** is like trying to find a needle in a haystack as large as a football field. The needle represents the **private key** and the haystack represents all the possible **private keys**.

In the same way, one-way functions are designed to be irreversible and cannot be decrypted.

What is a hash function?

Hashing is like a fingerprint for digital data. It is a process of taking a digital message and turning it into a fixed length code, which serves as a unique identifier.

Just like a fingerprint can identify a person, a hash can identify a digital message. Hashes are used in many applications, including ***Bitcoin*** transactions.